The evolution of visual versus auditory communication when produced with the same articulatory apparatus

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Gesture-first theories of language origins are based, in significant part, on the argument that gestures afford far more iconicity than vocalizations, which would have supported the emergence of early symbols (Goldin-Meadow, 2016; Tomasello, 2008). This claim is supported by semiotics experiments, which consistently demonstrate that gestures outperform non-linguistic vocalizations when people develop novel communication systems from scratch (e.g., Fay et al., 2014). This gestural advantage for iconicity is often attributed to the depictive potential of the visual modality compared to the auditory modality of speech, which is characterized as linear and ephemeral, and thus, highly limited in its potential for iconicity (e.g., Armstrong & Wilcox, 2007; Hockett, 1978). However, these comparisons of gesture and vocalization tend to conflate the perceptual modality of communication (visual versus auditory) with the specific physical properties of the articulators (e.g., two hands versus a larynx and vocal tract). Thus, it is not clear whether gestures afford more iconicity than vocalizations due to the perceptual modality of communication or to other differences between the articulators.

The current study seeks to tease apart the specific role of the perceptual modality in the affordance of iconicity by comparing the evolution of visual versus auditory communication when signals are produced by the same articulatory apparatus. We designed a novel interface—namely, dragging a mouse on a tablet—to "draw" a visual signal or an auditory one. For the generation of auditory signals, the tablet x- and y-axes control the frequency and timbre of the auditory output. Both auditory and visual signals are captured as 2D coordinate trajectories, providing a commensurable measurement for direct cross-modal

comparison. In the general experimental set-up, dyads of participants play an interactive communication game in which they take turns creating auditory or visual signals to refer to eight animal stimuli (see Fig. 1). Animals each have a distinct appearance, movement, and sound, displayed by five-second video clips. In each trial, the Sender uses the tablet to generate a signal for a target meaning within 30s. The Receiver guesses the target meaning from six alternatives, and their response times are recorded. After each trial, feedback is provided to both players on whether the guess was correct. The Sender and Receiver swap roles after each trial. Each participant cycles through five rounds, totaling 80 (2×5×8) trials per dyad.

Figure 1. The stimuli inventory of eight animal video screenshots. The upper and lower rows are respectively the visual and auditory signals sampled from the last round of two individuals.

Our results with 25 dyads ($N_{auditory}$=13, N_{visual}=12) showed that guessing accuracy was above chance from the start in both modalities, indicating successful communication (see Fig. 2 left). Dyads became more accurate over rounds. While performance in the visual modality was consistently higher than in the auditory modality, both improved at similar rates, suggesting similar learning patterns. As accuracy improved, guessing time decreased, reflecting adaptation across rounds, with visual signals processed faster than auditory signals. This advantage of visual modality appeared at the start and plateaued over rounds.

We next analyzed whether the signals in each modality showed evidence of conventionalization. Within individuals we tested whether the forms participants produced for each meaning became more stable over rounds. Within dyads, we tested whether participants converged on more similar forms for each meaning. For these analyses, we treated the mouse movement trajectories (in both modalities) as images, and employed VGG19, a convolutional neural network pretrained on the ImageNet dataset, to quantify the perceptual distance between images (see Fig. 2 right). Within dyads, the results yielded no decrease of perceptual distance over rounds, indicating a lack of convergence. However, within individuals, the perceptual distance between round *n* and *n-1* decreased in both modalities, indicating that individuals' signals became more stable in form. The lack of dyadic convergence might be attributed to the abundant distinctive features in the stimuli, imposing little pressure for dyads to align their signals.

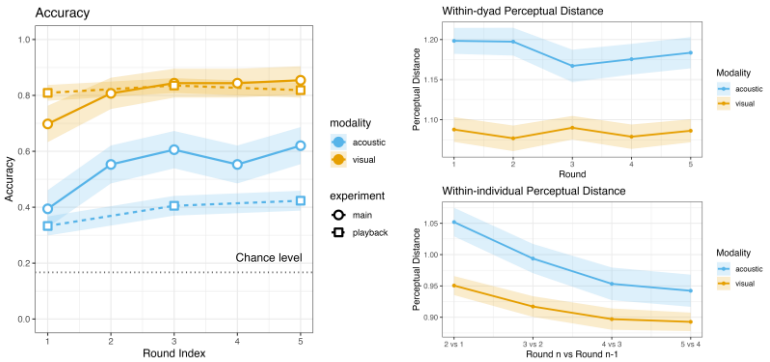

Figure 2. Accuracy across rounds between the two modalities in the main and playback experiment (left). While individuals stabilized on the form of signals, there is a lack of evidence showing dyadic convergence (right).

Our primary interest was in the affordance of iconicity, and we reasoned that improved accuracy after round one could result from feedback. To assess the iconicity of signals over rounds, we conducted a playback experiment with naïve participants. Participants ($N_{auditory}$=141, N_{visual}=146) were tasked to guess the animal from the signals, which were derived from the main experiment dataset, totaling 480 stimuli (2 modalities × 3 rounds (1, 3, 5) × 8 videos × 10 samples). Each signal received 8~13 datapoints of guessing. The auditory modality exhibited increased accuracy and marginally reduced response times, whereas the visual modality plateaued at high accuracy from the start (see Fig. 2 left). Notably, naïve participants underperformed in the auditory modality suggesting that the steeper accuracy gain in the main experiment was driven in part by learning effects rather than iconicity affordances alone.

Overall, our findings suggest that the visual modality affords more iconicity compared to the auditory modality. However, the physical apparatus maintained a confound in the modality comparison. Whereas the visual drawings remained continuously visible throughout each trial, the auditory signals were rapid-fading and required replay, creating a persistence–transience asymmetry that is not intrinsic to the modalities themselves. Ongoing work controls for this factor by making the drawings transient (a "trace" condition) to allow more direct comparison between the perceptual modalities independent of articulation.

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